Conditionals Coding QS

You said:

condition all concept with coding qs and sol

ChatGPT said:

Alright—let's build your **Conditionals (if-else)** concept from zero to advanced, with **coding questions + solutions in Java & Python**. This is super important for interviews and problem-solving.

🔥 1. What are Conditionals?

Conditionals are used to **make decisions in code**.

👉 Real life example:

- If it rains → take umbrella ☔
- Else → go normally

🧠 2. Types of Conditions

✅ (1) Simple `if`

👉 Runs only if condition is true

Python

```
Python
age = 18
if age >= 18:
    print("You can vote")
```

Java

```
Java
int age = 18;
if (age >= 18) {
    System.out.println("You can vote");
}
```

✅ (2) `if-else`

👉 One block runs if true, other if false

Python

```
Python
num = 5
if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Java

```
Java
int num = 5;
if (num % 2 == 0) {
    System.out.println("Even");
} else {
    System.out.println("Odd");
}
```

✅ (3) `if-elif-else` (Python) / `else-if` (Java)

👉 Multiple conditions

Python

```
Python
marks = 85

if marks >= 90:
    print("A")
elif marks >= 75:
    print("B")
else:
    print("C")
```

Java

```
Java
int marks = 85;

if (marks >= 90) {
    System.out.println("A");
} else if (marks >= 75) {
    System.out.println("B");
} else {
    System.out.println("C");
}
```

✅ (4) Nested if

👉 if inside another if

Python

```
Python
age = 20
if age >= 18:
    if age >= 21:
        print("Can drink")
```

Java

```
Java
int age = 20;

if (age >= 18) {
    if (age >= 21) {
        System.out.println("Can drink");
    }
}
```

✅ (5) Ternary Operator (Short if-else)

Python

```
Python
num = 10
result = "Even" if num % 2 == 0 else "Odd"
print(result)
```

Java

```
Java
int num = 10;
String result = (num % 2 == 0) ? "Even" : "Odd";
System.out.println(result);
```

⚡ 3. Important Condition Operators

Operator	Meaning
==	Equal
!=	Not equal
> <	Greater / Less
>= <=	Greater/less equal
&& / and	AND
`	

Operator	Meaning
`!` / `not`	NOT

💪 4. Practice Questions (With Solutions)

🔹 Q1: Check Positive, Negative, Zero

Python

```
Python
num = int(input())

if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

Java

```
Java
import java.util.*;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int num = sc.nextInt();

        if (num > 0) {
            System.out.println("Positive");
        } else if (num < 0) {
            System.out.println("Negative");
        } else {
            System.out.println("Zero");
        }
    }
}
```

🔹 Q2: Find Largest of 3 Numbers

Python

```
Python
a, b, c = 10, 20, 15

if a >= b and a >= c:
    print(a)
elif b >= a and b >= c:
    print(b)
else:
    print(c)
```

Java

```
Java
int a = 10, b = 20, c = 15;

if (a >= b && a >= c) {
    System.out.println(a);
} else if (b >= a && b >= c) {
    System.out.println(b);
} else {
    System.out.println(c);
}
```

🔹 Q3: Check Leap Year

Python

```
Python
year = 2024

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
else:
    print("Not Leap Year")
```

Java

```
Java
int year = 2024;

if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
    System.out.println("Leap Year");
} else {
    System.out.println("Not Leap Year");
}
```

◆ Q4: Grade Calculator

Python

```
Python
marks = 72

if marks >= 90:
    print("A")
elif marks >= 75:
    print("B")
elif marks >= 50:
    print("C")
else:
    print("Fail")
```

Java

```
Java
int marks = 72;

if (marks >= 90) {
    System.out.println("A");
} else if (marks >= 75) {
    System.out.println("B");
} else if (marks >= 50) {
    System.out.println("C");
} else {
    System.out.println("Fail");
}
```

🔹 Q5: Check Vowel or Consonant

Python

```
Python
ch = 'a'

if ch in 'aeiou':
    print("Vowel")
else:
    print("Consonant")
```

Java

```
Java
char ch = 'a';

if ("aeiou".indexOf(ch) != -1) {
    System.out.println("Vowel");
} else {
    System.out.println("Consonant");
}
```

🔹 Q6: Simple Calculator

Python

```
Python
a = 10
b = 5
op = '+'

if op == '+':
    print(a + b)
elif op == '-':
    print(a - b)
elif op == '*':
    print(a * b)
elif op == '/':
    print(a / b)
```

Java

```
Java
int a = 10, b = 5;
char op = '+';

if (op == '+') {
    System.out.println(a + b);
} else if (op == '-') {
    System.out.println(a - b);
} else if (op == '*') {
    System.out.println(a * b);
} else if (op == '/') {
    System.out.println(a / b);
}
```

🚀 5. Interview-Level Questions

Try these next:

1. Check if number is palindrome
2. Check Armstrong number
3. Electricity bill calculation
4. Income tax calculator
5. Triangle validity check
6. Login system (username & password check)

🎯 6. Pro Tips

- Always check **edge cases** (0, negative, large values)
- Avoid too many nested `if` → use clean logic
- Use **ternary** for short conditions
- Practice combining conditions (`AND`, `OR`)

🔥 Next Step (Recommended)

If you want, I can:
✅ Give **50 condition-based interview questions (with solutions)**
✅ Teach **pattern + logic building step-by-step**

✅ Move to **loops + conditions combo (very important for coding rounds)**

Just tell me:
👉 "Give 50 condition questions" or
👉 "Teach me next level (loops + conditions)"

You said:

Give 50 condition-based interview questions (with solutions)

ChatGPT said:

Got it — here are **50 condition-based interview questions with clean, optimized solutions (Python + Java)**.
I've grouped them **Easy → Medium → Hard** so you can learn step-by-step.

🟢 EASY (1–15)

1. Check Even or Odd

```
Python
num = 7
print("Even" if num % 2 == 0 else "Odd")
```

```
Java
int num = 7;
System.out.println(num % 2 == 0 ? "Even" : "Odd");
```

2. Positive / Negative / Zero

```
Python
num = -5
if num > 0: print("Positive")
elif num < 0: print("Negative")
else: print("Zero")
```

```
Java
int num = -5;
if (num > 0) System.out.println("Positive");
else if (num < 0) System.out.println("Negative");
else System.out.println("Zero");
```

3. Largest of Two

```
Python
a, b = 10, 20
print(max(a, b))
```

```
Java
int a = 10, b = 20;
System.out.println(Math.max(a, b));
```

4. Largest of Three

```
Python
a,b,c = 10,20,15
print(max(a,b,c))
```

```
Java
int a=10,b=20,c=15;
System.out.println(Math.max(a, Math.max(b,c)));
```

5. Check Leap Year

```
Python
y=2024
print("Leap" if (y%4==0 and y%100!=0) or y%400==0 else "Not")
```

```
Java
int y=2024;
System.out.println(((y%4==0 && y%100!=0)||
(y%400==0))?"Leap":"Not");
```

6. Divisible by 5 and 11

```
Python
n=55
print("Yes" if n%5==0 and n%11==0 else "No")
```

```
Java
int n=55;
System.out.println(n%5==0 && n%11==0 ? "Yes":"No");
```

7. Vowel or Consonant

```
Python
ch='a'
print("Vowel" if ch in "aeiou" else "Consonant")
```

```
Java
char ch='a';
System.out.println("aeiou".indexOf(ch)!=-1?"Vowel":"Consonant");
```

8. Alphabet / Digit / Special

```
Python
ch='@'
if ch.isalpha(): print("Alphabet")
elif ch.isdigit(): print("Digit")
else: print("Special")
```

```
Java
char ch='@';
if(Character.isLetter(ch)) System.out.println("Alphabet");
else if(Character.isDigit(ch)) System.out.println("Digit");
else System.out.println("Special");
```

9. Check Uppercase or Lowercase

```
Python
ch='A'
print("Upper" if ch.isupper() else "Lower")
```

```
Java
char ch='A';
System.out.println(Character.isUpperCase(ch)?"Upper":"Lower");
```

10. Valid Triangle

```
Python
a,b,c=3,4,5
print("Valid" if a+b>c and a+c>b and b+c>a else "Invalid")
```

```
Java
int a=3,b=4,c=5;
System.out.println((a+b>c && a+c>b &&
b+c>a)?"Valid":"Invalid");
```

11. Simple Calculator

```
Python
a,b,op=10,5,'+'
print(a+b if op=='+' else a-b if op=='-' else a*b if op=='*'
else a/b)
```

```
Java
int a=10,b=5; char op='+';
System.out.println(op=='+'?a+b:op=='-'?a-b:op=='*'?a*b:a/b);
```

12. Pass or Fail

```
Python
marks=35
print("Pass" if marks>=33 else "Fail")
```

```
Java
int marks=35;
System.out.println(marks>=33?"Pass":"Fail");
```

13. Grade System

```
Python
m=85
print("A" if m>=90 else "B" if m>=75 else "C" if m>=50 else
"F")
```

```
Java
int m=85;
System.out.println(m>=90?"A":m>=75?"B":m>=50?"C":"F");
```

14. Check Multiple of 3 or 7

```
Python
n=21
print("Yes" if n%3==0 or n%7==0 else "No")
```

```
Java
int n=21;
System.out.println(n%3==0 || n%7==0?"Yes":"No");
```

15. Absolute Value

```
Python
n=-10
print(abs(n))
```

```
Java
int n=-10;
System.out.println(Math.abs(n));
```

🟡 MEDIUM (16–35)

16. Swap Without Temp

```
Python
a,b=5,10
a,b=b,a
```

```
Java
int a=5,b=10;
a=a+b; b=a-b; a=a-b;
```

17. Check Armstrong

```
Python
n=153
print("Yes" if n==sum(int(d)**3 for d in str(n)) else "No")
```

```
Java
int n=153,temp=n,sum=0;
while(temp>0){int d=temp%10; sum+=d*d*d; temp/=10;}
System.out.println(sum==n?"Yes":"No");
```

18. Palindrome Number

```
Python
n=121
print("Yes" if str(n)==str(n)[::-1] else "No")
```

```
Java
int n=121,rev=0,temp=n;
while(temp>0){rev=rev*10+temp%10; temp/=10;}
System.out.println(rev==n?"Yes":"No");
```

19. Even & Divisible by 3

```
Python
n=12
print("Yes" if n%2==0 and n%3==0 else "No")
```

20. Profit or Loss

```
Python
cp,sp=100,120
print("Profit" if sp>cp else "Loss")
```

21. Electricity Bill (basic slab)

```
Python
u=120
bill=u*5 if u<=100 else 100*5+(u-100)*10
print(bill)
```

22. Age Group

```
Python
age=25
print("Child" if age<13 else "Teen" if age<20 else "Adult")
```

23. Find Middle of 3

```
Python
a,b,c=10,20,15
print(sorted([a,b,c])[1])
```

24. Check Prime

```
Python
n=7
print("Prime" if n>1 and all(n%i for i in range(2,n)) else
"Not")
```

25. Divisible by 2,3,5

```
Python
n=30
print("Yes" if n%2==0 and n%3==0 and n%5==0 else "No")
```

26. Triangle Type

```
Python
a,b,c=3,3,3
print("Equilateral" if a==b==c else "Isosceles" if a==b or
b==c or a==c else "Scalene")
```

27. Character ASCII Category

```
Python
ch='A'
print("Upper" if 'A'<=ch<='Z' else "Lower" if 'a'<=ch<='z'
else "Other")
```

28. Discount Calculation

```
Python
price=1000
print(price*0.9 if price>500 else price)
```

29. Login System

```
Python
u,p="admin","1234"
print("Login" if u=="admin" and p=="1234" else "Fail")
```

30. Time Greeting

```
Python
h=14
print("Morning" if h<12 else "Afternoon" if h<18 else "Night")
```

31. Check Power of 2

```
Python
n=8
print("Yes" if n>0 and (n&(n-1))==0 else "No")
```

32. BMI Category

```
Python
bmi=22
print("Normal" if 18.5<=bmi<=24.9 else "Other")
```

33. Quadratic Roots Type

```
Python
a,b,c=1,2,1
d=b*b-4*a*c
print("Real" if d>=0 else "Imaginary")
```

34. Check Perfect Square

```
Python
n=16
print("Yes" if int(n**0.5)**2==n else "No")
```

35. Compare 3 Numbers

```
Python
a,b,c=5,5,5
print("Equal" if a==b==c else "Not Equal")
```

🔴 HARD (36–50)

36. Income Tax (slab)

```
Python
income=700000
tax=0
if income>500000:
    tax+=(income-500000)*0.2
    income=500000
if income>250000:
    tax+=(income-250000)*0.1
print(tax)
```

37. ATM Withdrawal

```
Python
amt=500; bal=1000
print("Success" if amt%100==0 and amt<=bal else "Fail")
```

38. Strong Number

```
Python
import math
n=145
print("Yes" if n==sum(math.factorial(int(d)) for d in str(n))
else "No")
```

39. Harshad Number

```
Python
n=18
print("Yes" if n%sum(map(int,str(n)))==0 else "No")
```

40. Duck Number

```
Python
n="1023"
print("Yes" if '0' in n[1:] else "No")
```

41. Automorphic Number

```
Python
n=25
print("Yes" if str(n*n).endswith(str(n)) else "No")
```

42. Neon Number

```
Python
n=9
print("Yes" if sum(map(int,str(n*n)))==n else "No")
```

43. Spy Number

```
Python
n=1124
digits=list(map(int,str(n)))
print("Yes" if
sum(digits)==__import__('functools').reduce(lambda
x,y:x*y,digits) else "No")
```

44. Magic Number (simple)

```
Python
n=19
while n>9:
    n=sum(map(int,str(n)))
print("Magic" if n==1 else "Not")
```

45. Perfect Number

```
Python
n=6
print("Yes" if sum(i for i in range(1,n) if n%i==0)==n else
"No")
```

46. LCM Condition

```
Python
a,b=4,6
import math
print(math.lcm(a,b))
```

47. GCD Condition

```
Python
import math
print(math.gcd(12,18))
```

48. Temperature Category

```
Python
t=30
print("Hot" if t>25 else "Cold")
```

49. Voting Eligibility with ID

```
Python
age,id=True
print("Eligible" if age>=18 and id else "Not")
```

50. Password Strength

```
Python
p="Abc@123"
print("Strong" if len(p)>=6 and any(c.isdigit() for c in p)
and any(c.isupper() for c in p) else "Weak")
```

🚀 What Next?

If you want to level up:

👉 I can give:

- 🔥 **50 loop + condition problems (MOST IMPORTANT for interviews)**
- 🧠 **Pattern + logic building training (step-by-step)**

- 💻 **DSA-level problems (Amazon / Google style)**

Just say:
👉 *"Next level problems"* or
👉 *"Teach me logic building step-by-step"*

You said:

Give me 50 loop + condition problems (MOST IMPORTANT for interviews)

ChatGPT said:

Good—this is the **MOST IMPORTANT section for coding interviews**.
Loops + Conditions = 🔥 Core of problem solving.

I'll give you **50 problems with clean logic + optimized solutions (Python + Java)**
👉 Structured as **Easy → Medium → Hard (interview level)**

🟢 EASY (1–15)

1. Print numbers 1 to N

```
Python
for i in range(1, 11):
    print(i)
```

```
Java
for(int i=1;i<=10;i++) System.out.println(i);
```

2. Print even numbers till N

```
Python
for i in range(1,11):
    if i%2==0: print(i)
```

```
Java
for(int i=1;i<=10;i++){
    if(i%2==0) System.out.println(i);
}
```

3. Sum of first N numbers

```
Python
n=10
s=0
for i in range(1,n+1): s+=i
print(s)
```

```
Java
int n=10,sum=0;
for(int i=1;i<=n;i++) sum+=i;
System.out.println(sum);
```

4. Factorial

```
Python
n=5
f=1
for i in range(1,n+1): f*=i
print(f)
```

```
Java
int n=5,f=1;
for(int i=1;i<=n;i++) f*=i;
System.out.println(f);
```

5. Multiplication table

```
Python
n=5
for i in range(1,11):
    print(n*i)
```

```
Java
int n=5;
for(int i=1;i<=10;i++)
    System.out.println(n*i);
```

6. Count digits

```
Python
n=1234
count=0
while n>0:
    count+=1
    n//=10
print(count)
```

7. Reverse number

```
Python
n=123
rev=0
while n>0:
    rev=rev*10+n%10
    n//=10
print(rev)
```

8. Palindrome number

```
Python
n=121
print("Yes" if str(n)==str(n)[::-1] else "No")
```

9. Sum of digits

```
Python
n=123
print(sum(map(int,str(n))))
```

10. Count even digits

```
Python
n=1234
print(sum(1 for d in str(n) if int(d)%2==0))
```

11. Print digits

```
Python
n=123
for d in str(n): print(d)
```

12. Check Armstrong (3-digit)

```
Python
n=153
print("Yes" if n==sum(int(d)**3 for d in str(n)) else "No")
```

13. Find largest digit

```
Python
n=927
print(max(map(int,str(n))))
```

14. Count odd numbers till N

```
Python
count=0
for i in range(1,11):
    if i%2: count+=1
print(count)
```

15. Sum of even numbers

```
Python
s=0
for i in range(1,11):
    if i%2==0: s+=i
print(s)
```

🟡 MEDIUM (16–35)

16. Prime check

```
Python
n=7
print("Prime" if n>1 and all(n%i for i in range(2,n)) else
"Not")
```

17. Print primes till N

```
Python
for n in range(2,20):
    if all(n%i for i in range(2,n)):
        print(n)
```

18. Fibonacci series

```
Python
a,b=0,1
for _ in range(10):
    print(a)
    a,b=b,a+b
```

19. GCD (loop)

```
Python
a,b=12,18
while b:
    a,b=b,a%b
print(a)
```

20. LCM using loop

```
Python
a,b=4,6
l=max(a,b)
while True:
    if l%a==0 and l%b==0:
        print(l)
        break
    l+=1
```

21. Perfect number

```
Python
n=6
print("Yes" if sum(i for i in range(1,n) if n%i==0)==n else
"No")
```

22. Strong number

```
Python
import math
n=145
print("Yes" if n==sum(math.factorial(int(d)) for d in str(n))
else "No")
```

23. Sum of factorial series

```
Python
import math
s=0
for i in range(1,6):
    s+=math.factorial(i)
print(s)
```

24. Count primes in range

```
Python
count=0
for n in range(2,20):
    if all(n%i for i in range(2,n)):
        count+=1
print(count)
```

25. Reverse string

```
Python
s="hello"
print(s[::-1])
```

26. Palindrome string

```
Python
s="madam"
print("Yes" if s==s[::-1] else "No")
```

27. Count vowels

```
Python
s="hello"
print(sum(1 for c in s if c in "aeiou"))
```

28. Remove duplicates

```
Python
s="aabbcc"
print("".join(dict.fromkeys(s)))
```

29. Pattern (star triangle)

```
Python
for i in range(1,5):
    print("*"*i)
```

30. Inverted triangle

```
Python
for i in range(5,0,-1):
    print("*"*i)
```

31. Number pattern

```
Python
for i in range(1,5):
    for j in range(1,i+1):
        print(j,end="")
    print()
```

32. Sum of odd numbers

```
Python
print(sum(i for i in range(1,11) if i%2))
```

33. Count digits divisible by 3

```
Python
n=123456
print(sum(1 for d in str(n) if int(d)%3==0))
```

34. Product of digits

```
Python
n=123
p=1
for d in str(n): p*=int(d)
print(p)
```

35. Sum of squares

```
Python
print(sum(i*i for i in range(1,6)))
```

🔴 HARD (36–50)

36. Print Armstrong numbers in range

```
Python
for n in range(100,500):
    if n==sum(int(d)**3 for d in str(n)):
        print(n)
```

37. Check Harshad number

```
Python
n=18
print("Yes" if n%sum(map(int,str(n)))==0 else "No")
```

38. Check Automorphic

```
Python
n=25
print("Yes" if str(n*n).endswith(str(n)) else "No")
```

39. Check Neon

```
Python
n=9
print("Yes" if sum(map(int,str(n*n)))==n else "No")
```

40. Spy number

```
Python
n=1124
digits=list(map(int,str(n)))
from functools import reduce
print("Yes" if sum(digits)==reduce(lambda x,y:x*y,digits) else
"No")
```

41. Magic number

```
Python
n=19
while n>9:
    n=sum(map(int,str(n)))
print("Magic" if n==1 else "Not")
```

42. Frequency of digits

```
Python
n=122333
from collections import Counter
print(Counter(str(n)))
```

43. Longest word

```
Python
s="I love coding"
print(max(s.split(), key=len))
```

44. Count words

```
Python
s="hello world python"
print(len(s.split()))
```

45. Toggle case

```
Python
s="AbC"
print("".join(c.lower() if c.isupper() else c.upper() for c in
s))
```

46. Remove spaces

```
Python
s="a b c"
print(s.replace(" ",""))
```

47. Find duplicates

```
Python
s="aabbcc"
print([c for c in set(s) if s.count(c)>1])
```

48. Second largest digit

```
Python
n=98321
digits=sorted(set(map(int,str(n))))
print(digits[-2])
```

49. Sum of alternate digits

```
Python
n="12345"
print(sum(int(n[i]) for i in range(0,len(n),2)))
```

50. Pattern (pyramid)

```
Python
for i in range(1,5):
    print(" "*(5-i)+"*"*(2*i-1))
```

🚀 What Next (VERY IMPORTANT)

Now you are ready for:

👉 🔥 **DSA Patterns (Sliding Window, Two Pointer, Hashing)**
👉 💻 **Top 100 Interview Questions (Amazon, Google level)**
👉 🧠 **Step-by-step logic building (like a pro coder)**

Just say:

👉 **"Teach me DSA patterns"**
👉 **"Give me top interview questions"**
👉 **"Make me 0 → Advanced coder roadmap"**

You said:

Loop alll concept explain wit coding qs and ans

ChatGPT said:

Alright—let's build **LOOPS from zero → advanced** in a way that actually sticks for interviews.

🔥 1. What is a Loop?

A loop is used to **repeat a block of code multiple times**.

👉 Example: print numbers 1–100 without writing 100 lines.

🧠 2. Types of Loops

✅ (1) `for` Loop (Most Used)

👉 Used when **you know how many times to run**

Python

```
Python
for i in range(1, 6):
    print(i)
```

Java

```
Java
for(int i=1; i<=5; i++){
    System.out.println(i);
}
```

✅ (2) `while` Loop

👉 Used when **condition-based looping**

Python

```
Python
i = 1
while i <= 5:
    print(i)
    i += 1
```

Java

```
Java
int i = 1;
while(i <= 5){
    System.out.println(i);
    i++;
}
```

✅ (3) `do-while` (Java only)

👉 Runs **at least once**

```
Java
int i = 1;
do{
    System.out.println(i);
    i++;
}while(i <= 5);
```

⚡ 3. Loop Control Statements

🔹 `break` → stop loop

```
Python
for i in range(1,10):
    if i == 5:
        break
    print(i)
```

🔹 `continue` → skip iteration

```
Python
for i in range(1,6):
    if i == 3:
        continue
    print(i)
```

🔹 `pass` (Python only)

```
Python
for i in range(5):
    pass
```

🧩 4. Important Loop Patterns

⭐ Pattern 1: Counting

```
Python
for i in range(1,11):
    print(i)
```

⭐ Pattern 2: Accumulator (Sum/Product)

```
Python
s = 0
for i in range(1,6):
    s += i
print(s)
```

⭐ Pattern 3: Digit Extraction

```
Python
n = 123
while n > 0:
    d = n % 10
    print(d)
    n //= 10
```

⭐ Pattern 4: Nested Loops (Patterns)

```
Python
for i in range(1,4):
    for j in range(1,4):
        print("*", end="")
    print()
```

💪 5. Practice Questions (With Solutions)

🔹 Q1: Sum of N numbers

```
Python
n=5
print(sum(range(1,n+1)))
```

```
Java
int n=5,sum=0;
for(int i=1;i<=n;i++) sum+=i;
System.out.println(sum);
```

🔹 Q2: Factorial

```
Python
n=5
f=1
for i in range(1,n+1): f*=i
print(f)
```

```
Java
int n=5,f=1;
for(int i=1;i<=n;i++) f*=i;
System.out.println(f);
```

Q3: Reverse Number

```
Python
n=123
rev=0
while n>0:
    rev=rev*10+n%10
    n//=10
print(rev)
```

```
Java
int n=123,rev=0;
while(n>0){
    rev=rev*10+n%10;
    n/=10;
}
System.out.println(rev);
```

Q4: Palindrome Number

```
Python
n=121
print("Yes" if str(n)==str(n)[::-1] else "No")
```

```
Java
int n=121,rev=0,temp=n;
while(temp>0){
    rev=rev*10+temp%10;
    temp/=10;
}
System.out.println(rev==n?"Yes":"No");
```

Q5: Prime Number

```
Python
n=7
print("Prime" if n>1 and all(n%i for i in range(2,n)) else
"Not")
```

```
Java
int n=7; boolean prime=true;
for(int i=2;i<n;i++){
    if(n%i==0) prime=false;
}
System.out.println(prime?"Prime":"Not");
```

◆ Q6: Fibonacci Series

```
Python
a,b=0,1
for _ in range(10):
    print(a)
    a,b=b,a+b
```

```
Java
int a=0,b=1;
for(int i=0;i<10;i++){
    System.out.println(a);
    int temp=a+b;
    a=b;
    b=temp;
}
```

◆ Q7: Multiplication Table

```
Python
n=5
for i in range(1,11):
    print(n*i)
```

```
Java
int n=5;
for(int i=1;i<=10;i++)
    System.out.println(n*i);
```

🔹 Q8: Count Digits

```
Python
n=1234
count=0
while n>0:
    count+=1
    n//=10
print(count)
```

```
Java
int n=1234,count=0;
while(n>0){
    count++;
    n/=10;
}
System.out.println(count);
```

🔹 Q9: Sum of Digits

```
Python
n=123
print(sum(map(int,str(n))))
```

```
Java
int n=123,sum=0;
while(n>0){
    sum+=n%10;
    n/=10;
}
System.out.println(sum);
```

🔹 Q10: Pattern (Triangle)

```
Python
for i in range(1,5):
    print("*"*i)
```

```
Java
for(int i=1;i<=4;i++){
    for(int j=1;j<=i;j++) System.out.print("*");
    System.out.println();
}
```

🧠 6. Interview Tips

👉 Use **for loop** → when range known
👉 Use **while loop** → when condition unknown
👉 Use **nested loops** → patterns & matrices
👉 Optimize loops (avoid unnecessary iterations)

⚠️ Common Mistakes

❌ Infinite loop
❌ Wrong loop condition
❌ Off-by-one error (`i <= n` vs `i < n`)
❌ Forget increment/decrement

🚀 Next Level

If you want to go PRO:

👉 I can teach:

- 🔥 Loop + Array DSA patterns (MOST IMPORTANT)
- 💻 Sliding Window / Two Pointer
- 🧠 How to solve ANY problem step-by-step

Just say:

👉 **"Teach me DSA with loops"**
👉 **"Give me real interview problems"**

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